

PHILIPP BENJAMIN ARETZ

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RESEARCH INTERESTS

Effective field theory and precision QCD; factorization and resummation for collider observables; medium effects in heavy-ion collisions; mathematical structure of quantum field theory.

EDUCATION

PhD in Physics, Massachusetts Institute of Technology, Cambridge, MA, USA 2024–present
MASt in Mathematics (Part III), University of Cambridge, Cambridge, UK 2023–2024
Pass with Distinction, 92% — ranked 9th of 294
BSc in Physics, Heidelberg University, Heidelberg, Germany 2020–2023
1.0 (best possible)
Thesis: *The Keldysh Functional Integral and Bounds on Truncated Expectation Values* (advisor: Prof. Manfred Salmhofer)
Abitur, Goethe-Schule Bochum, Bochum, Germany 2020
1.0

RESEARCH POSITIONS

Research Assistant, Heidelberg University, Heidelberg, Germany 2022–2023
CRC/TRR 191 Symplectic Structures in Geometry, Algebra and Dynamics
Differential delay equations: periodic delay orbits and smooth orbit families. Magnetic billiards: KAM-based extension near convex boundaries.
Tutor, Heidelberg University, Heidelberg, Germany 2021–2022

PUBLICATIONS AND PREPRINTS

Published

P. B. Aretz, M. Salmhofer, *A rigorous Keldysh functional integral for fermions*, J.Statist.Phys. 193 (2026) 28, arXiv:2508.01787 [math-ph].
P. Albers, **P. Aretz**, I. Seifert, *Families of periodic delay orbits*, J. Differential Equations 410 (2024) 228-250, arXiv:2304.07550 [math].

Preprints

P. B. Aretz, K. Lee, S. T. Schindler, I. W. Stewart, *Factorization of elastic, single, and double diffractive $\$pp\$ scattering$* , arXiv:2607.07788 [hep-ph].

CONTRIBUTED CONFERENCE TALKS

Diffraction at the LHC, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna 2026

<i>Generalised Detectors in Heavy Ion Collisions</i> , ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna	2026
<i>Diffraction Across Colliders</i> , TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder	2026
<i>Diffraction Across Colliders</i> , SCET 2026, Seoul, South Korea	2026

Internal seminars

Generalised Detector Observables in the Heavy Ion Environment, MIT EFT Group Meeting (2025)	
Resurgence and Renormalons in QCD, MIT EFT Group Meeting (2025)	
Generalised Detector Observables in the Heavy Ion Environment, MIT Heavy Ions Group Meeting (2025)	

CONFERENCES AND SCHOOLS ATTENDED

ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Erwin Schrödinger Institute - Vienna (Poster)	2026
TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder, CO, USA	2026
SCET 2026, Seoul, South Korea (Talk)	2026
GGI School on Theory of Fundamental Interactions, Florence, Italy	2026
SCET 2025, Tucson, AZ, USA	2025

AWARDS AND FUNDING

Tom Frank Fellowship , Massachusetts Institute of Technology	2024
Jennings Prize , University of Cambridge	2024
DAAD Scholarship , Deutscher Akademischer Austauschdienst	2023
Wolfson College Scholarship , Wolfson College, Cambridge	2023
e-fellows.net Scholarship	

TEACHING

Classical Mechanics 8.01(TA) , MIT	Fall
Classical Mechanics 8.01(TA) , MIT	Fall

SERVICE AND LEADERSHIP

Treasurer , Edgerton House, MIT	2025–2026
President MIT Rowing Club , MIT Rowing Club	2026–present
Organiser, First Year Seminar , Massachusetts Institute of Technology	2024–2025
Buddy for first-year students , Heidelberg University	2021–2022

TECHNICAL SKILLS

Python (NumPy, pandas, TensorFlow); C++ (ROOT, Pythia); Mathematica; MATLAB; LaTeX.

LANGUAGES

German (native); English (C2, Cambridge Grade A, TOEFL 117/120); French (basic); Spanish (basic); Latin (Latinum).